

Adarsh Yadav

AI Engineer | GenAI · LLM · RAG Systems

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SUMMARY

AI Engineer building production-grade **Large Language Model (LLM)** systems — **Retrieval-Augmented Generation (RAG)**, multi-agent workflows, and scalable GenAI backends — with a B.Tech from IIT Roorkee. Skilled in **prompt engineering**, **LLM evaluation**, **guardrails**, and **vector search**, and experienced in deploying event-driven serverless APIs on Azure. Strong CS fundamentals backed by 700+ DSA problems solved.

EDUCATION

Indian Institute of Technology Roorkee

B.Tech

2022 – 2026

CGPA: 7.99 / 10

EXPERIENCE

FnMathLogic Consulting Services

June 2026 – Present

GenAI-powered Education Platform

- **Engineered LLM-powered content generation pipelines** for automated, grade and subject-specific daily lesson planning, incorporating contextual data, prompt engineering, and structured output generation.
- **Scaled high-volume LLM workloads** through Python multiprocessing, multithreading, and concurrent API execution, improving generation throughput and minimizing sequential inference bottlenecks.
- **Architected and deployed event-driven serverless APIs** with **Azure Functions** and **Azure Storage**, achieving **99.9% production uptime** while supporting scalable lesson-generation workloads.

PROJECTS

DB Explorer — MCP Server for AI-Powered Database Exploration 

Python, MCP SDK, SQLAlchemy (AsyncIO), aiosqlite, asyncpg

- **Built a 10-tool MCP server** following the **Model Context Protocol** standard, enabling Claude AI to explore schemas, run queries, and analyze SQL databases through **natural language via JSON-RPC over STDIO**.
- **Implemented read-only SQL execution with safety guardrails** — **regex-based write detection**, query timeouts, row limits, and multi-statement blocking to prevent data modification by the AI agent.
- **Designed async database layer** using **SQLAlchemy AsyncIO** with support for **SQLite, PostgreSQL, and MySQL**; built schema introspection, ERD generation, query plan analysis, and per-column data quality statistics.

Multi-Agent Banking Chatbot 

Python, FastAPI, PostgreSQL, Redis, Groq (LLaMA 3.1), React

- **Built** a multi-agent banking assistant where a **coordinator LLM classifies user intent** and routes to specialist agents (account, transaction, service), each with scoped DB context injected into the system prompt; deployed full-stack on **Render + Vercel + Neon + Upstash**.
- **Implemented a PII masking pipeline** that replaces account numbers, PAN, Aadhaar, phone, and email with tokens before any data reaches the LLM, and unmask the reply before returning to the user.
- **Designed JWT authentication** with bcrypt hashing, **access + refresh tokens**, Redis-backed **rate limiting** (20 req/min per user), and a **confirm-before-write pattern** for all DB mutations.
- **Added observability endpoint** tracking per-agent latency, call counts, and process CPU/memory; structured as a **coordinator → specialist agent** pattern that is easy to extend with new agents.

SKILLS

Languages: Python, C++, SQL, JavaScript

AI / LLM Engineering: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Multi-Agent Systems, Model Context Protocol (MCP), LangGraph, LLM Evaluation & Benchmarks, LLM Guardrails, Embeddings, Sentence Transformers, Vector Search (pgvector, k-NN)

Backend & Systems: REST APIs, FastAPI, Node.js, Express.js, System Design, Scalability, Microservices, Serverless (Azure Functions), Authentication, Async Programming

Databases & Caching: PostgreSQL, MongoDB, Redis, pgvector

Cloud & DevOps: Microsoft Azure AI, Docker, CI/CD, Git, GitHub, Linux, Observability (Structlog)

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

ACHIEVEMENTS

- Secured **AIR 9162** in JEE Advanced 2022 among 1.6 lakh+ qualified candidates.
- Solved **1000+ algorithmic problems** across Codeforces, CodeChef, and LeetCode, reflecting strong problem-solving and DSA fundamentals.